

Machine Learning Project: Autonomous Vision and Image Classification

This technical report details the development, training, and empirical evaluation of a deep Convolutional Neural Network (CNN) pipeline designed for real-time visual classification.

Technical Architecture and Pipeline:

1. **Model Architecture:** Implemented a deep ResNet-50 residual network backbone initialized with ImageNet pre-trained weights. Custom classification heads were added with batch normalization and dropout regularization ($p=0.4$).
2. **Dataset and Preprocessing:** Utilized the ISIC 2020 dataset comprising 33,126 high-resolution dermoscopic images. Applied comprehensive data augmentation pipelines including random affine rotations, horizontal flipping, and Gaussian color jitter to prevent model overfitting.
3. **Loss Optimization and Training:** Addressed severe class imbalance by implementing focal loss with $\gamma=2.0$ and alpha weighting. The network was optimized using AdamW with cosine annealing learning rate scheduler over 50 epochs on dual NVIDIA GPUs.
4. **Quantitative Evaluation:** Evaluated on a 20% held-out test split, achieving an Area Under ROC Curve (AUC-ROC) of 0.934, sensitivity of 89.1%, and precision of 86.7%.
5. **Model Deployment:** Exported the trained model weights to ONNX and TensorRT runtime engines, achieving an inference latency of under 18 milliseconds per image frame.